

# docker\_class\_7(50)

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## DOCKER HUB

we can store the image in docker hub and anyone can access the image from docker hub same as git hub like docker hub

in the docker components docker registry is the last components

docker hub is the default registry and docker registry is used to store the images

in the docker registry there are two types

1. cloud based registry --> docker hub, GCR google container Registry , Amazon ECR elastic container registry
2. local registry --> nexus, jfrog, DTR docker trusted registry

```
git clone https://github.com/Harishjangidi/staticsite-docker.git
```

```
cd statistics-docker
```

```
cd html
```

```
vim Dockerfile
```

```
FROM nginx
```

```
COPY . /usr/share/nginx/html
```

```
docker build -t image1 .
```

if we want to push to docker hub let see the steps:

login to dockerhub

goto repos --> create repo --> where you can syntax also how to push --> click on create

## Pushing images

You can push a new image to this repository using the CLI:

```
docker tag local-image:tagname new-repo:tagname
```

```
docker push new-repo:tagname
```

after click on create it will give command to push i.e

To push a new tag to this repository

```
docker push jangidiharish07924/statistics-docker:tagname
```

- 1.create image
- 2.tag the image
- 3.push the image

### Tag

docker tag <local-image>:<tag> <dockerhub-username>/<repo-name>:<tag>

### docker login

### Push

docker push <dockerhub-username>/<repo-name>:<tag>

in our case :

docker login

docker tag test-stats:latest jangidiharish07924/test-stats-docker:latest

docker push jangidiharish07924/test-stats-docker:latest

**\*\*WE HAVE TO TAKE THE CODE FROM DEVELOPERS**

**\*\*\*WRITE A DOCKER FILE AS PER THE SOURCE CODE**

**\*\*\*BUILD THE DOCKER FILE THEN IMAGE WILL GETS CREATED**

**\*\*PUSH THE IMAGE TO DOCKER HUB**

- TAG THE IMAGE
- LOGIN INTO DOCKER HUB FROM SERVER (docker login)
- PUSH THE IMAGE

we will have different types of docker file as like nodejs,javascript,python.

devops will focus automation

now we will setup jenkins and push to dockerhub

setup jenkins is done

create a job name: docker pipeline1

1.

```
pipeline {
  agent any

  stages {
    stage("Code") {
      steps {
        git "https://github.com/devops0014/staticsite-docker.git"
      }
    }
  }
}
```

2.

```
pipeline {  
  agent any  
  
  stages {  
  
    stage("Code") {  
      steps {  
        git "https://github.com/devops0014/staticsite-docker.git"  
      }  
    }  
  
    stage("Build") {  
      steps {  
        sh 'docker build -t pavan:one ./html'  
      }  
    }  
  }  
}
```

save and build

check console it should be success

now you can this command: docker images

pavan image will be there

3.

```
pipeline {  
  agent any  
  
  stages {  
  
    stage("Code") {  
      steps {  
        git "https://github.com/devops0014/staticsite-docker.git"  
      }  
    }  
  
    stage("Build") {  
      steps {  
        sh 'docker build -t pavan:one ./html'  
      }  
    }  
  }  
}
```

```

    }
stage("Deploy") {
  steps {
    sh '''
    docker rm -f cont1 || true
    docker run -d --name cont1 -p 8082:80 pavan:one
    '''
  }
}
}
}
}
}

```

copy ip adress with port in browser you can able to access.

4.  
while going build i want to give input name so we have to use build with parameter  
setup (parameters UI + pipeline) to select imagename and container name while at build time

```

pipeline {
  agent any

  stages {

    stage("Code") {
      steps {
        git "https://github.com/devops0014/staticsite-docker.git"
      }
    }

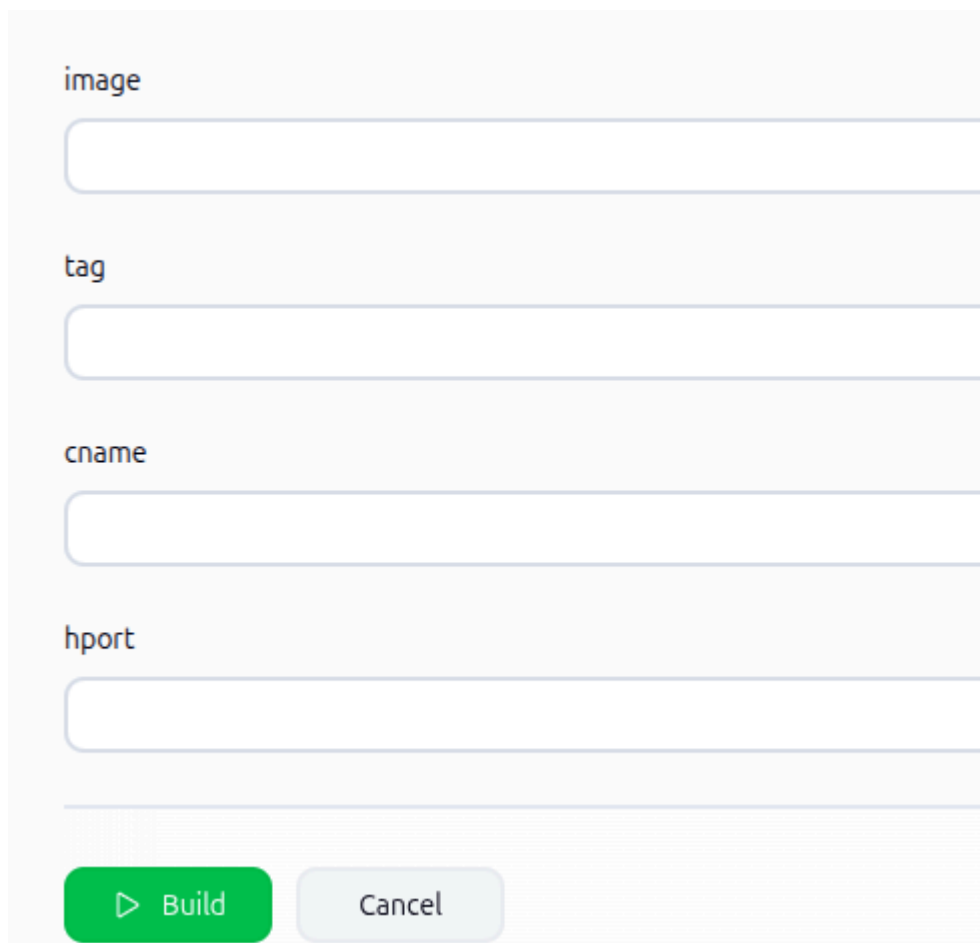
    stage("Build") {
      steps {
        sh 'docker build -t ${image}:${tag} ./html'
      }
    }

    stage("Deploy") {
      steps {
        sh 'docker run -d --name ${cname} -p ${hport}:80 ${image}:${tag}'
      }
    }

  }
}

```

save and build with parameters



A screenshot of the Jenkins Docker build configuration form. It features four input fields labeled 'image', 'tag', 'cname', and 'hport' arranged vertically. At the bottom, there are two buttons: a green 'Build' button with a play icon and a light blue 'Cancel' button.

TASK: when ever the build is done the image we have to push docker hub  
i want to see the list of iamges and list containers in jenkins so for that small config need to do

1.docker plugin needs to install

2.manage jenkins-->docker

3. Docker integration with jenkins

vim /lib/systemd/system/docker.service

goto line 19 th upto -H keep after that tcp:0.0.0.0:4243 -H unix:///var/run/docker.sock

save and continue

systemctl daemon reload

restart docker

4. manage jenkins-->clouds

new cloud-->mydocker-->type(docker)-->create

add cloud details-->dockerhosturl:tcp://0.0.0.0:4243 click test connection -->save

5. managejenkins-->docker-->mydocker: here we can see list of containers

Note: only we can see running containers